

CSE 203 Computer Ethics and Cyber Law – Tutorial 1 Mark 20

Consider the following scenario:

“Faria is a second-year CSE student at Netrokona University. She has disability on her left hand which is weak from her birth. Her family is not solvent enough to support her education, so she has to look for an alternate income source. As a computer science student, she came up with the idea of making a mobile application for tuition seekers. Many university students and guardians made accounts in her app. Later a shop came to visit her and offered her a handsome amount of money to share app users’ personal information so they could use it for personalized business advertisements. Faria badly needed the money and could not refuse the offer.”

1. According to “Cyber Security Act-2023”, analyze the above scenario and (i) identify the rules that have been broken (if any) by Ms. Faria and (ii) the corresponding punishments she will be subject to. [7 marks]
2. Evaluate the act of Ms. Faria using (i) Rule Utilitarianism and (ii) Social Contract ethical theories. [7 marks]
3. Differentiate among morals, ethics, and virtue. What are the expected features of a good corporate code of ethics? [6 marks]

CSE 203 Computer Ethics and Cyber Law – Tutorial 2 Mark 20

1. “XYZ University requires that all computers connected to the campus network run up-to-date antivirus software. When a student’s computer is discovered to have a virus, its network connection is cut until a staff member can remove the virus. If the computer is not running up-to-date antivirus software, the student is fined \$100.”
- Is this a morally justifiable policy? Use the Social Contract Theory. *[5 marks]*
2. Classify and compare perpetrators of computer crimes. *[5 marks]*
3. Differentiate between Symmetric key and Public key encryptions. *[5 marks]*
4. Should you make a software trademark, copyright it, or claim for a patent? Justify your answer. *[5 marks]*

CSE 203 Computer Ethics and Cyber Law – Tutorial 3 Mark 20

- Q1. Explain a few common ethical issues regarding the relationship between IT professionals and Suppliers. Discuss a few initiatives that will support the ethical practices of IT users. [5 marks]
- Q2. "Rifat and his friends, aged between 13 and 17, were very fond of action-oriented, role-playing computer games. After some time, severe changes were noticed in their behavior. They wanted to play the games for longer durations. They became very aggressive, started using offensive language, and often got into conflicts with their peers on minor issues. A psychiatrist suggested that the violence that the children observed in the computer games had a negative effect on them."
Do you think it is ethically correct for a programmer or a company to produce such games? Answer taking software engineering code of ethics into consideration. [5 marks]
- Q3. Explain the following terms with respect to computer ethics: (i) Globalization (ii) Digital Divide (iii) Net Neutrality. [6 marks]
- Q4. Discuss the dilemma between moral responsibility and personal/organizational risks of whistle-blowing. [4 marks]

1. I was not in the class last week due to

[5]

- (a) I was late to rise
- (b) I have completed all the chapters
- (c) I am not interested in electronics but BCS
- (d) Statement is false

Given the following figure, answer the following questions

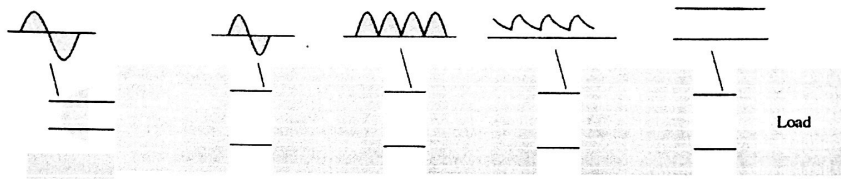


Figure 1: DC power

2. What will be the input voltage in the Bangladesh context?

- (a) 210
- (b) 440
- (c) 220
- (d) 110

3. First two blocks are

- (a) Filter and Regulator
- (b) Regulator and Filter
- (c) Any combination can fit
- (d) None is true

4. Find the name of the last block.

[2]

- (a) Filter
- (b) Regulator
- (c) Transformer
- (d) None is true

5. The top part of the figure states

[2]

- (a) Input voltage shape for each block
- (b) Output voltage shape after each block
- (c) Both are true
- (d) None is true

6. If the load voltage should be 12V, what should be the first block output voltage? [2]

- (a) 12V
- (b) $>12V$
- (c) $\gg 12V$
- (d) $<12V$

7. Ripple factor for the full wave is

- (a) 121 %
- (b) 131 %
- (c) 148 %
- (d) None is true

8. Diode rating in rectifier depends on

- (a) Filter resistance
- (b) Filter Capacitance
- (c) Load resistance
- (d) Load Capacitance

Given the following figure 2, answer the following questions

9. What is the leftmost block's name?

- (a) Reference
- (b) Co reference
- (c) Reference voltage
- (d) None of these

10. What type of regulator block it is?

- (a) Shunt voltage

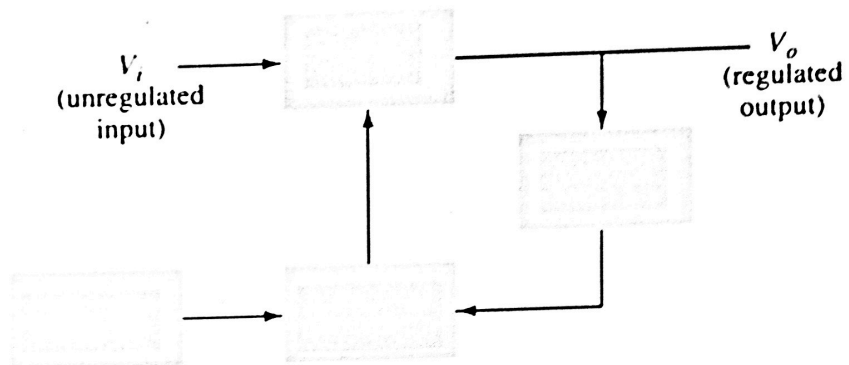


Figure 2: Regulated power supply block diagram

- (b) Series voltage
 - (c) Series current
 - (d) Shunt current
11. What device is used for reference voltage?
- (a) Diode
 - (b) LED
 - (c) Transistor
 - (d) None of these
12. Is it possible to develop this regulated power supply with op-amp?
- (a) True
 - (b) False

Jahangirnagar University

Department of Computer Science and Engineering

2nd Year 1st Semester B.Sc. (Hons.)

Course Title: Numerical Methods; Course No: CSE-205; Class Test: 02

Total Marks: 20

Fill in the Blanks [Marks: 10]

1. The process of eliminating variables in the Gauss elimination method is also known as _____ elimination.
2. The Gauss elimination method works by transforming the coefficient matrix into a _____.
3. In partial pivoting, the row with the largest _____ in the current column is swapped with the pivot row.
4. A matrix is invertible if and only if its determinant is _____.
5. The Jacobi and Gauss-Seidel methods are classified as _____ methods for solving linear systems.
6. The Gauss-Seidel method updates the values _____ in each iteration.
7. The Gauss-Seidel method generally converges _____ than the Jacobi method.
8. In linear interpolation, the estimated value lies between two known values and follows the equation of a _____.
9. Lagrange interpolation is a _____ interpolation technique used for curve fitting based on given discrete data points.
10. Interpolation is a technique used to estimate values of a function at _____ data points within a given range.

Answer the following question [Marks: 5+5]

Question 1:

A game developer is working on a physics-based simulation where an object's movement requires computing the square root of non-integer values for speed calculations. The game's performance depends on quick approximations instead of expensive square root computations.

The developer has a lookup table for square roots of integers:

x	$\sqrt{x}$
1	1
2	1.4142
3	1.7321
4	2
5	2.2361

To approximate the square root of 3.5 for the game's physics engine, they decide to use **linear interpolation**. Calculate the estimated value and explain why interpolation is useful in computer graphics.

Question 2:

An electrical engineer is working on optimizing power distribution in a city's power grid. The system of linear equations representing the power flow is large.

- The Jacobi method is initially chosen because it allows parallel computation of updates for different nodes.
- However, it is observed that the method takes many iterations to converge.
- The engineer considers switching to the Gauss-Seidel method to speed up convergence.

Which method should the engineer use for faster results, and why?

Department of Computer Science and Engineering, JU
2nd Year 1st Semester B.Sc. (Hons.)
Course No: CSE-205 ; Class Test: 03, Total Marks: 20

Answer all the following questions

Question 1 [7 Marks]:

A drone navigation system relies on accurate estimations of angular velocities to adjust its orientation. The onboard computer estimates the rate of change of the angle θ , where $\theta(t)=\sin(t)$, with t in radians, at a particular moment when $t=0.45$ seconds.

To compute the numerical derivative at increasing values of h from 0.01 to 0.04 with a step size of 0.005. Analyze the variation of error in each step.

Question 2 [6 Marks]:

A space research team is simulating the motion of a satellite re-entering Earth's atmosphere. The total heat generated on the satellite's surface is modeled by a complex function involving altitude, velocity, and varying atmospheric density:

$$Q(t) = \int_0^T f(t, v(t), h(t)) dt$$

Tasks:

- ✓ **Explain** why traditional methods of calculus (analytical integration) are not feasible for computing the total heat $Q(t)$ in this scenario.
- ✓ **Discuss** at least one advantage and one trade-off of using numerical integration in real-world engineering problems.

Question 3 [7 Marks]:

An engineer is designing a sensor that measures the exponential growth of voltage across a capacitor in a circuit, where the voltage at time x seconds is modeled by the function $V(x)=e^x$. To estimate the total voltage accumulated over a time interval from $x=-1$ to $x=+1$ seconds, the engineer decides to use the **Composite Trapezoidal Rule** due to its computational simplicity.

To compute the approximate integral:

$$\int_{-1}^1 e^x dx$$

using:

- (i) $n=2$ equally spaced time intervals.
- (ii) $n=4$ equally spaced time intervals.



Department of CSE
Jahangirnagar University
Savar, Dhaka-1342.

Course No: Math 201

Class Test-02

Time: 40 Minutes

Marks: 30

1. Using Cauchy's residue theorem evaluate $\int_0^{\infty} \frac{dx}{x^4 + 16}$.
 2. If $\underline{a} = x^2y\hat{i} - 2xz\hat{j} + 2yz\hat{k}$, find $\underline{\nabla} \cdot (\underline{\nabla} \times \underline{a})$ at the point $(1, -1, 1)$.
 3. Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $\hat{i} - \hat{j} - 2\hat{k}$.
- A Show that $\underline{\nabla} \cdot \left(\frac{\underline{r}}{r^3} \right) = 0$, where $\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$.



Department of CSE
Jahangirnagar University
Savar, Dhaka-1342.

Course No: Math 201

Class Test-01

Time: 45 Minutes

Marks: 30

1. (a) Prove that the function $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic.
(b) Find a function v such that $f(z) = u + iv$ is analytic.
2. Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{z^2 - 4z + 3} dz$; where C is the circle (a) $|z| = 5$, (b) $|z| = 2$.
3. Expand $f(z) = \frac{1}{(z-1)(2-z)}$ in a Laurent series valid for
(a) $0 < z < 1$
(b) $0 < |z-2| < 1$.

Tutorial-02
Department of Computer Science and Engineering
Jahangirnagar University

Course Code: CSE-209

Time: 40 Minutes

Marks: 24

Answer any 4 of the following questions

1. Consider the following array elements.

Data	78	50	30	8	65	22	28	4	6	12	2	13
Index	1	2	3	4	5	6	7	8	9	10	11	12

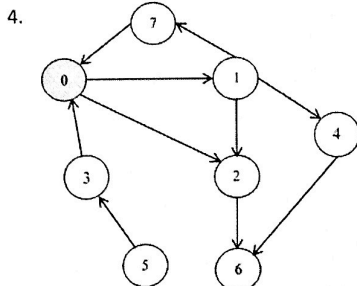
- a) Draw the Heap from the array, does the following array represent a *Max-Heap* with root at index 1?
 b) If not at what should the index to call the *Max-Heapify* function and call as many time of *Max-Heapify* to make it a Max Heap. If yes, explain why it is a Max-Heap.
2. Can you use the Master theorem to solve the following recurrences?
 a) $T(n) = 4T(n/2) + n$
 b) $T(n) = 4T(n-2) + cn$
 c) $T(N) = 6 * T(N/5) + N^3$

If yes, solve it with this method (give the value for ϵ where needed and use limit theorem for $O(\theta/\Omega)$. If no, show why you cannot use it.

3. Consider this recursive function **test(N)**:

```
int test(int N){
    if (N <= 4) return 2;
    int total = 0;
    for(int r = 1; r < N; r++){
        total = total + test(r);
    }
    return total;
}
```

- a) Write the recurrence formula for this function (including the base cases) for $N \geq 0$.
 b) Draw the tree that shows the function calls performed in order to compute **test(7)** (Hint. the root will be **test(7)** and it will have a child for each recursive call.)
 c) Compute the result value returned by **test(6)**.



Consider the left graph and answer the followings:

- a) Traverse this graph with Depth First Search (DFS) starting from node 0. If a vertex has more than one neighbors you must visit them in increasing order of vertex number, for example from node 1 you would visit the neighbors in order: 2,4,7.

Order from first to last	Visited vertex	Parent
1 st	0	-
2 nd		

- b) Label the edges as one of these categories: T (tree), B (backward), C (cross) or F(forward).

5. You are given the following adjacency list representation of a undirected graph.

0: 1, 2, 3, 7
 1: 0, 2, 4, 7
 2: 0, 1, 6
 3: 0, 5
 4: 1, 6
 5: 3
 6: 2, 4
 7: 0, 1

- a) Now construct the graph.
 b) Traverse the graph below in Breadth-First order, starting from vertex 2. Whenever you examine neighbors of a vertex, process the nodes according to the adjacency list (from the smallest to largest vertex number). Fill in the table (the vertices, edges and the distance).

Order from first to last edges added to the BFS tree	Vertex, v, (in order visited by BFS)	Edge: u, v (edge that adds v to the BFS tree)	Distance (as number of edges) from the source
1 st	2		
2 nd			
...			

Tutorial-03
Department of Computer Science and Engineering
Jahangirnagar University

Course Code: CSE-209

Time: 40 Minutes

Marks: 24

Answer any 4 of the following questions

1) You are given the following table

Value	A	B	C	D	E	F
Frequency	5	25	7	15	4	12

- Find the optimal strategy for encoding using Huffman Coding.
- Write down the encoding for the word "BED".

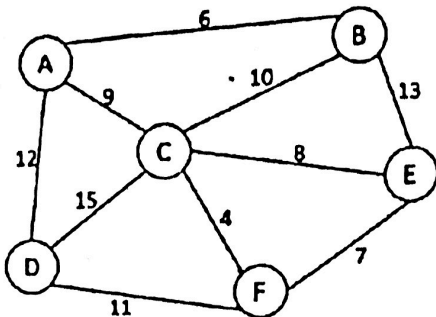
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2

2) A list of different activities with starting and ending times. Find the maximum size of non-overlapping activities that can be performed.

6

Activities	1	2	3	4	5	6	7	8
Start Time	1	0	1	4	2	5	3	4
Finish Time	3	4	2	6	9	8	5	5

3)



Consider the left graph and answer the followings:

- Construct a minimum spanning tree starting from node E using Prim's Algorithm.

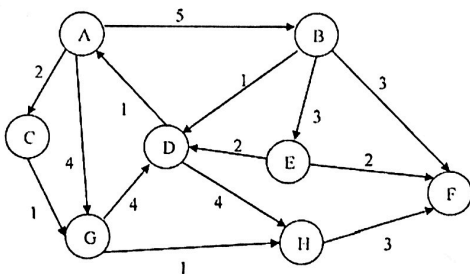
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Order from first to last	Distance	Path
A	0	-
...	...	...

2

- Draw the final tree.

4)



Consider the left graph and answer the followings:

- Use Dijkstra's algorithm with start node A to find the cost of the shortest path from A to each other node. For each node, record the cost and the previous node in the path. Keep track of the order you mark each node.

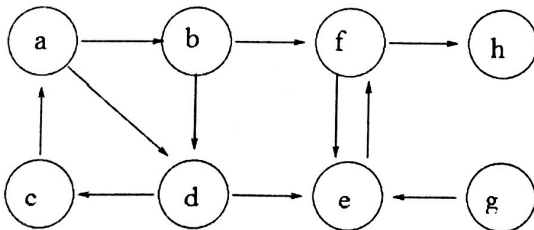
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Order from first to last	Cost	Parent
A	0	-
...	...	...

2

- Write a pseudocode to print the shortest cost path from node A to node F?

5)



6

Find the SCC for the given graph.

- Write down the four key steps for determining SCC for a given graph.
- Demonstrate step by step procedure.